Section 2 Group 4

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Paper Reading and Discussion Summary (In-class Activity)

We have chosen the paper entitled "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale" by Dosovitskiy, Alexey, et al. to read and discuss: https://arxiv.org/pdf/2010.11929

Dosovitskiy, Alexey, et al. "An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale." ArXiv:2010.11929 [Cs], 22 Oct. 2020, arxiv.org/abs/2010.11929, https://doi.org/10.48550/arXiv.2010.11929.

Title of the paper: An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Your names: Michael Reizenstein, Sam Moll, Dylan Shaffer

Your summary should answer the following questions:

1. **What** – What problem does the paper address and what is the main idea?

The goal of the paper is to explore how Transformers can be applied to image recognition.

The main idea is that a pure transformer applied directly to the sequence patches within an image can be just as effective as using convolutional networks to classify an image. Reliance on CNNs is not always necessary for computer vision tasks.

2. **Why** – Why is this problem important and why is the proposed approach useful?

One of the issues found in training traditional image classification models is that the convolutional layers are expensive to compute and in turn lead to relatively large training times. The proposed approach was one solution to this issue as transforming individual patches is less computational expensive as well as easy to analyze after. Compare their largest models, ViT-H/14 and ViT-L/16, to state-of-the-art CNNs The smaller ViT-L/16 model pre-trained on JFT-300M outperforms BiT-L on all tasks, while also requiring far less computational resources to train.

How – How does the method work (model, data, and evaluation)?

<u>Method:</u>

- They took subsections, or "patches" of the larger images and pass them into a transformer model as you would with tokens or words in an NLP application.

<u>Models:</u>

- Traditional "state-of-the-art CNNs"
- ViT-L/16 (other smaller/less trained models)

<u>Data:</u>

- The researchers used the ILSVRC-2012 ImageNet dataset with 1k classes and 1.3M images, which they call ImageNet, so look at the scalability of the model.

<u>Evaluation:</u>

- There were two main factors that the scientists looked to optimize for, and these were the raw performance of the ViT-L/16 versus CNNs and the other factor was how long it took to train the models.

Summary:

The main purpose of this paper was to explore the idea that alternative methods of image classification could be just as accurate as state of the art CNNs and that these new methods could lead to faster training times for models. The researchers found that the answer to both of these questions was yes, and the model they created performed just as well as modern CNN all while having been trained with less computational resources. This was done through the use of transformers on 16x16 pixel “patches” of an image. This process would then create attention (highlights the main feature/focus of the image) within the image and this drastically decreased training time as these images were now able to be processed faster into a usable form of data to be trained on. The way the data was processed led to it being treated more as a “token” from a word in a natural language processing model rather than a traditional image classification model.